

MLflow Experiment Tracking Report

Course: MLOps
Assignment: Observable ML with MLflow
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1. Introduction

This report demonstrates the use of MLflow to track and analyze machine learning experiments. A PyTorch neural network was trained on the MNIST dataset while logging parameters, metrics, and artifacts using MLflow. The purpose of this assignment is to move from silent training to observable machine learning workflows where experiments can be compared and reproduced.

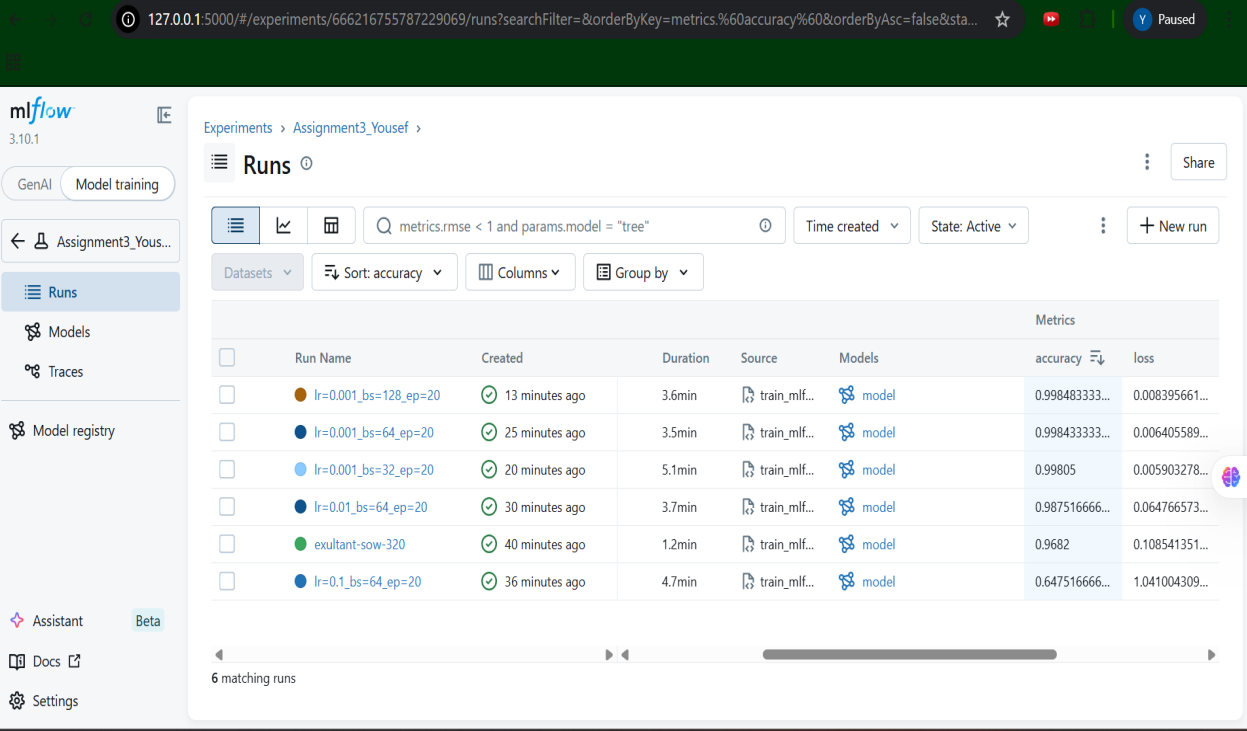
2. Experimental Setup

The model used in this assignment is a simple feed-forward neural network implemented in PyTorch. The architecture includes a flatten layer followed by two fully connected layers with a ReLU activation function. The model was trained on the MNIST dataset using cross-entropy loss and the Adam optimizer. MLflow was used to track key hyperparameters including learning rate, batch size, and number of epochs, while also logging training accuracy and loss for each experiment run.

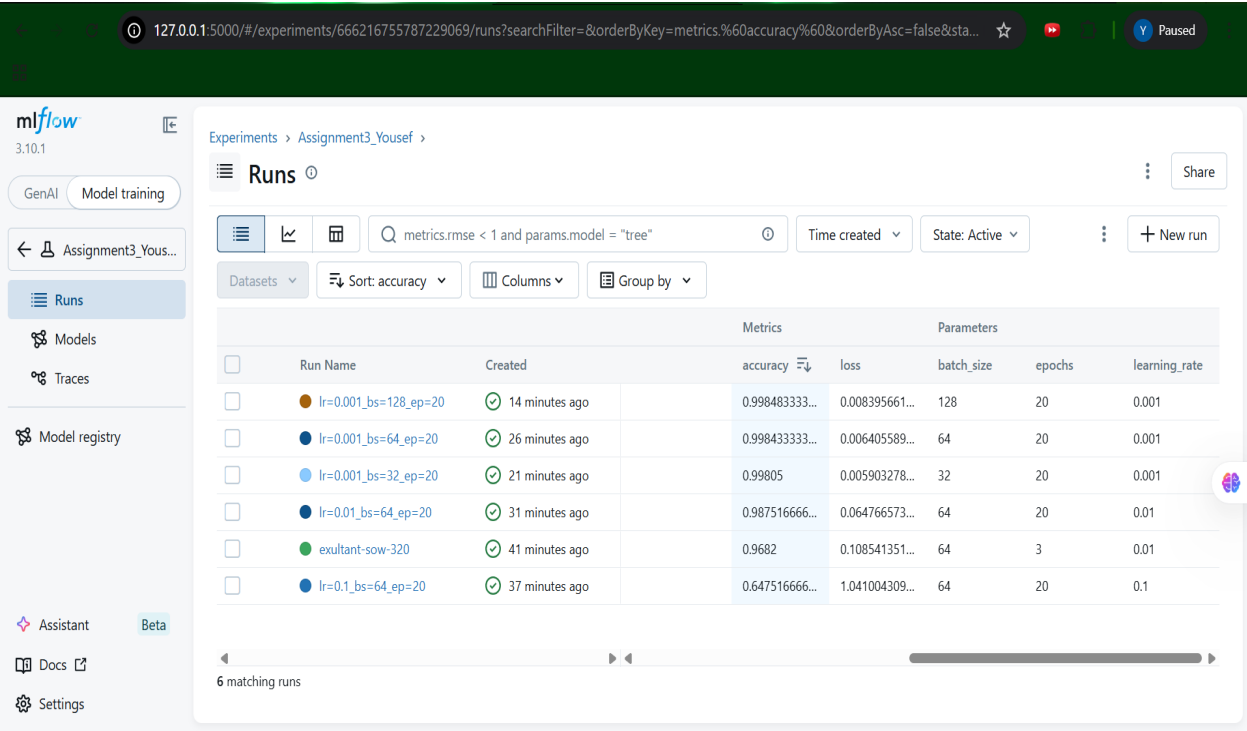
3. Experiment Tracking Results

The MLflow experiment dashboard provides a table view of all experiment runs including their hyperparameters and resulting metrics. This allows easy comparison of different configurations and identification of the best performing run.

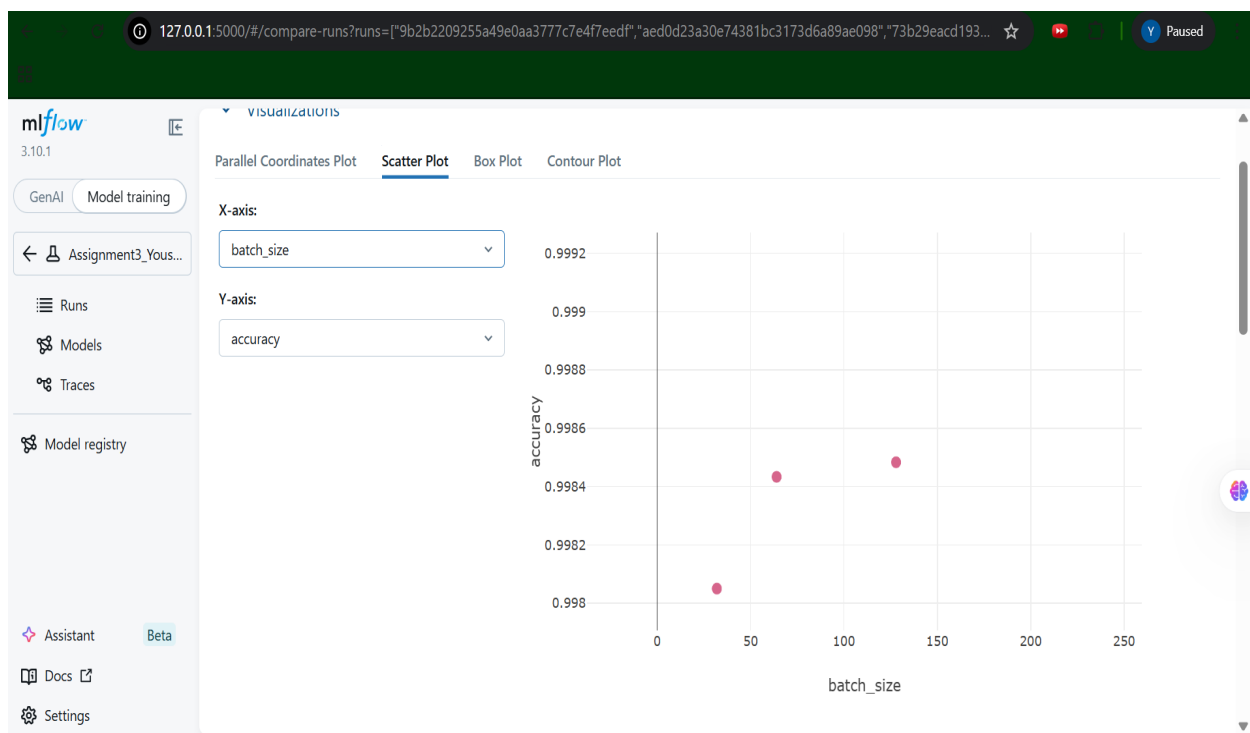
Table View of All MLflow Runs Sorted by Accuracy



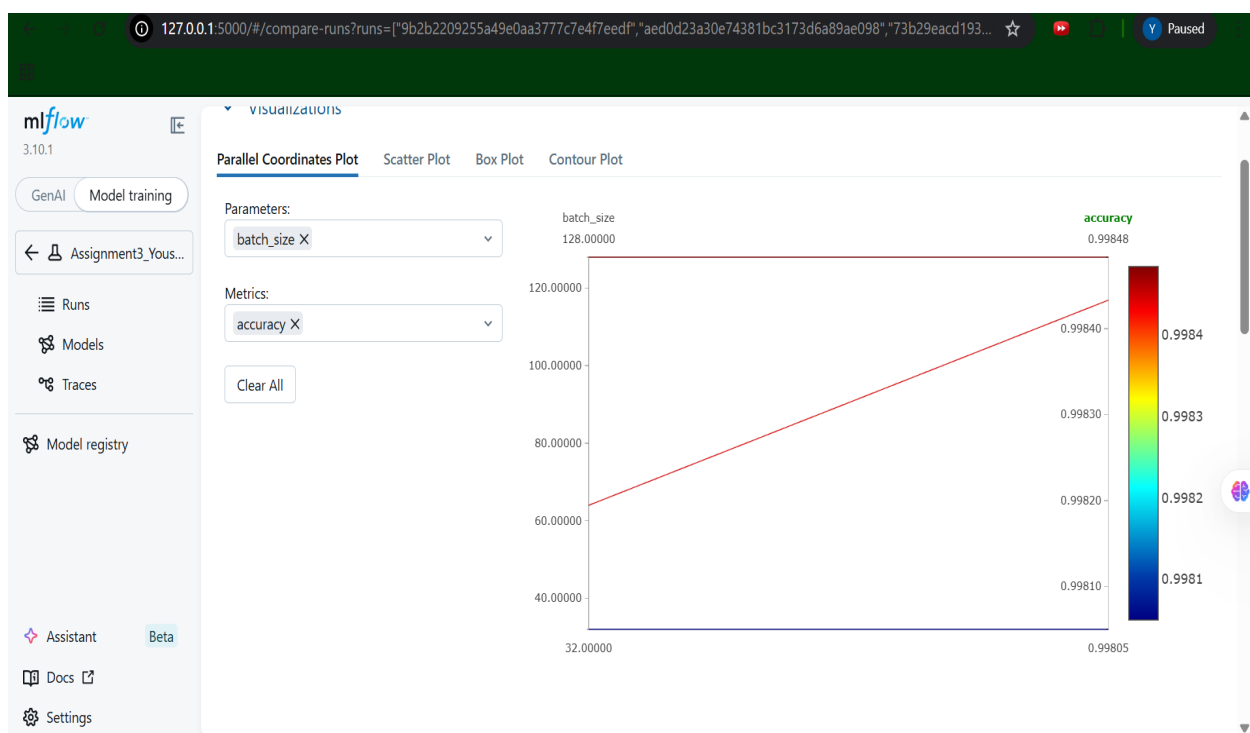
Detailed Run Parameters and Metrics in MLflow



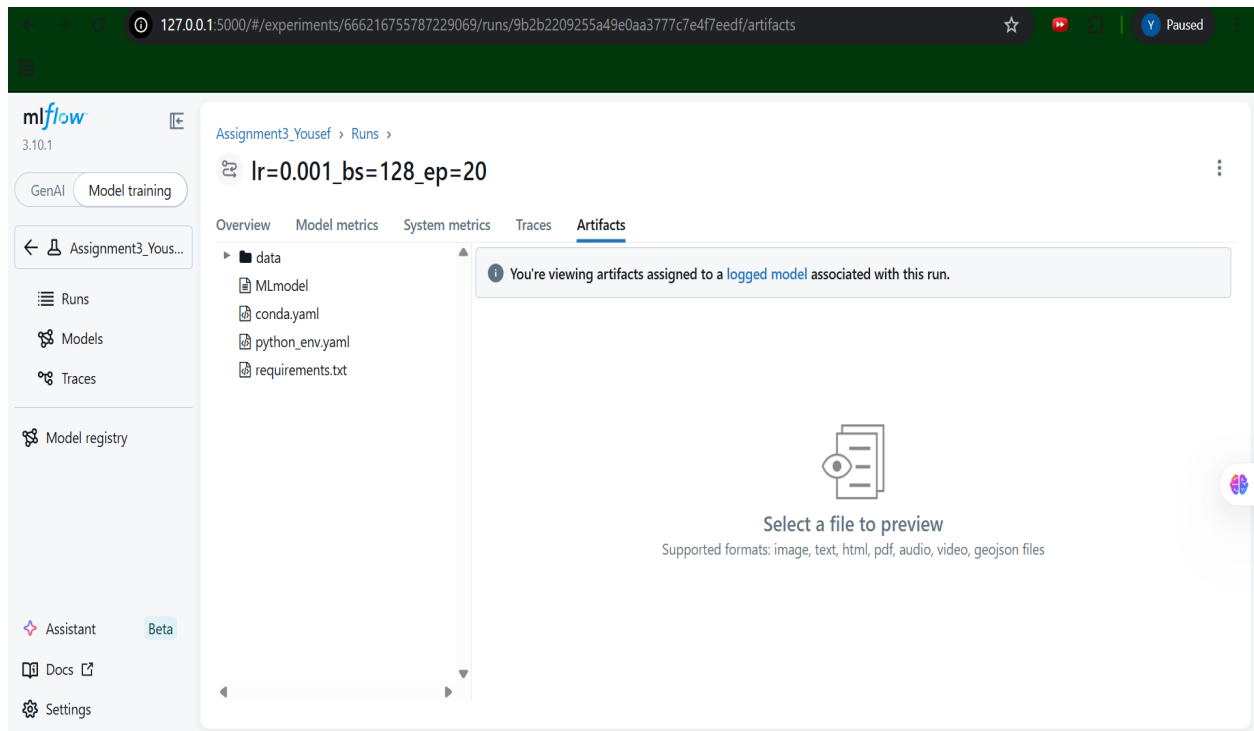
Scatter Plot Comparison of Batch Size vs Accuracy



Parallel Coordinates Plot Showing Parameter Influence



MLflow Artifact Gallery with Saved Model and Environment Files



4. Analysis of Results

The results show that lower learning rates produced more stable training and higher accuracy values. In particular, the configuration with `learning_rate = 0.001` and `batch_size = 128` achieved the best performance with accuracy close to 0.998. Conversely, the run using a high learning rate of 0.1 resulted in unstable training and significantly lower accuracy. Batch size variations also slightly influenced model performance, with larger batch sizes generally providing slightly more stable convergence in this experiment.

5. Conclusion

This assignment demonstrates the usefulness of MLflow for experiment tracking and model reproducibility. By logging parameters, metrics, and artifacts, MLflow allows researchers to systematically compare experiments and identify optimal hyperparameter configurations. The experiment results clearly show how different hyperparameters influence training behavior and final model performance.